

```
{
s[i] <- sum((as.integer(abs(diff(created))))<t)/500
i=i+1
}
return(s)
}
```

To create a cumulative value of the arriving tweets within a given interval, `countarrival()` uses `sum()` function over `diff()` function after converting the values into an integer.

```
>s <- countarrival(created)
>x<-seq(1,15,1)
>y<-s
>lo<- loess(y~x)
>plot(x,y)
>lines(predict(lo), col='red', lwd=2)

# sum((as.integer(abs(diff(created))))<t)/500
```

To have a smooth time series curve, the `loess()` function has been used with the `predict()` function. Predicted values based on the linear regression model, as provided by `loess()`, are plotted along with the x-y frequency values.

This is a classic example of probability distribution

of arrival probabilities. The pattern in Figure 5 shows a cumulative Poisson Distribution, and can be used to model the number of events occurring within a given time interval. The X-axis contains one-second time intervals. Since this is a cumulative probability plot, the likelihood of the next tweet arriving corresponds to the X-axis value or less than that. For instance, since 4 on the X-axis approximately corresponds to 60 per cent on the Y-axis, the next tweet will arrive in 4 seconds or less than that time interval. In conclusion, we can say that all the events are mutually independent and occur at a known and constant rate per unit time interval.

This data analysis and visualisation shows that the arrival pattern is random and follows the Poisson Distribution. The reader may test the arrival pattern with a different keyword too. 

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